FINANCING EQUITY IN EDUCATION: COMPARATIVE STUDY OF FUNDING MODELS ACROSS COUNTRIES

Jhanavi Sankar

Education financing is central to expanding access, enhancing socio-economic mobility, and promoting equity within school systems globally. Education spending constitutes an investment, as demonstrated by research indicating that each additional year of schooling increases annual earnings by approximately 9-10%.[1]The sources of financing—such as general taxes, property taxes, sales taxes, donations, cash grants, and subsidies—have significant implications for the structure of educational systems.[2] Careful consideration of both the source and allocation of funding is essential to ensure the selection of appropriate mechanisms for public education financing. In many countries, particularly developing nations, the design of financing mechanisms plays a critical role in determining whether children from disadvantaged backgrounds can access quality education comparable to their more advantaged peers.[3]

The international community has recognized the benefits of investing in education, including a 40% reduction in poverty since the 1980s and strengthened workforce development, which has led to the adoption of the Incheon Declaration.[4] This was a move towards allocating at least 4-6% of a country's GDP to education spending. Decades of reform, however, have not solved the crisis of educational financing in developing countries. Weak fiscal capacity exacerbated by debt pressures, declining global aid to education, and public budgets being constrained and unevenly distributed result in inadequate financing.[5] New analysis by UNESCO states that aid is expected to decline further, by over 25% by 2027. 17% to half of public education spending in low-income countries relies on aid.[6] These trends are particularly concerning given the rising out-of-school population and widening learning gaps at the elementary and middle-school levels.[7] Consequently, it is essential to utilize available funds efficiently and pursue innovative approaches to address these disparities.

This study contributes to these debates by examining and comparing education financing models across countries, with a focus on how different mechanisms influence equity. By analyzing traditional and innovative approaches, this study aims to identify key lessons for policymakers and international organizations working to design equitable financing mechanisms for public education.

CONCEPTUAL FRAMEWORK: MODELS OF EDUCATION FINANCING AND EQUITY

Education Financing is a socio-political decision-making process that involves policy and legal frameworks, administration and governance structures that regulate how these funds are collected, allocated, and distributed.[8] The Central/Federal Government of a country distributes these funds to different regions, groups, and types of educational institutions. Equity in education

financing is a central concern for policymakers, as it recognizes that students have different needs and lived experiences, and that providing identical inputs does not lead to fair opportunities. Robert Berne and Leanna Stiefle developed three equity principle concepts — equal opportunity, horizontal equity, and vertical equity.[9] These distinctions are not just theoretical but reflect policy debates and actively shaped funding designs for education. Equal opportunity is typically discussed in the context of financial resources, with variables such as ethnicity and race, where it is often observed that different areas within a school district may receive different levels of funding.[10] Horizontal equity or the equal treatment of equals suggests treating students with similar circumstances equally.[11] However, there is a challenge with defining equals as this principle assumes a level of similarity among students that may not exist in settings with diverse backgrounds, needs, and circumstances. By assuming similar students have the same starting point, it fails to account for personal differences and system level factors that create disparities. However, this could be a valid criterion to discuss resource distribution at the school level — differences in general education spending and funding for special education. The general education provides an equal base for all students, whereas the other funding could be used differentially across students.[12] Vertical equity is an important equity concept and is concerned with placing unequal burdens on persons of unequal circumstances.[13] In the school system, it involves providing additional support to students facing greater disadvantages based on income level, race, linguistic barriers, ethnicity, and so on.

A growing body of evidence suggests that when financing is conducted effectively and equitably in school systems, academic outcomes increase, and opportunity gaps reduce.[14] This is especially true for students hailing from low-income and marginalized communities who see improvements in graduation rates, higher education enrollments, employment, and wages. Prioritizing education financing can result in a higher skilled workforce, yield greater economic returns, and reduce inequalities in countries. To this end, education financing must be adequate to ensure systems have necessary resources, efficient to ensure maximum utilization of resources, and equitable to ensure quality education for all learners.[15] Education finance design based on principles of vertical equity, adequacy, and efficiency is needed now more than ever due to the learning crisis that several countries are facing. Millions of children are struggling with foundational literacy and numeracy skills amid a teacher shortage across many school systems worldwide. Countries have adopted diverse approaches to ameliorate this crisis, from centralized redistribution to weighted student funding.[16] Examining these models offer critical insights into how systems can translate resources into equitable opportunities for learners.

Education in developing countries is typically financed through Government budgets, household expenditures, international aid from organizations such as the World Bank and UNESCO, and innovative mechanisms such as impact bonds to bridge gaps.[17] When funding is distributed at the school district level, there is a mix of responsibilities with respect to funding allocation which concerns factors like resource categories, conditions of allocation, mechanisms of funding, and

amount of funding to each school.[18] Different resource categories such as infrastructure and operational costs may be held by different authorities. Typically, in developing countries, sub-national governments are responsible for final allocation to schools, however, central authorities may specify for what purpose the money is spent.[19] In terms of mechanisms for funding allocation, there is often a debate between the main allocation mechanism (e.g.: per-pupil formula) where schools have more flexibility and between other mechanisms that favor central reserves which allows governments to respond proactively during crises.[20] Predominantly four main bases are used by most countries to determine the extent of funding for each school. These are administrative discretion, incremental costs, bidding and bargaining, and formula funding.[21] Administrative discretion is based on an individual assessment of school needs and incremental costs considering historical expenditure to allocate funding. Bidding involves schools responding to open competition for additional funding such as a specialized programme. Formula funding involves allocation based on a predetermined formula that considers factors like student enrollment, subgroups, resources, and program needs.[22] None of these methods are used exclusively, usually, a combination of two or more methods is applied to determine funding. Information from OECD and partner countries states that formulas are commonly used for funding teaching staff whereas administrative discretion is utilized to fund capital expenditure.[23]

On average, countries spend about 3-5% of their GDP on financing public education and this includes roughly 10-20% of their total government expenditure.[24] According to the Education Finance Watch Report of 2024, overall education spending has increased over the past decade, but funding in low-income countries (LIC) remains a significant challenge. The amount a country spends on education per child is a measure of whether sufficient resources are allocated to public education. In a low-income country, expenditures per child per year are between $50 and $80.[25] In an upper middle-income country it is $1100 and in a high-income country, $8500. Furthermore, countries with resources are not spending them efficiently; for instance, South Africa, an upper-middle-income country, invests about $2400 per child per year. However, it has a severe learning poverty rate of 79%, meaning 10-year-olds are unable to read basic texts with comprehension.[26] This discussion is not only a question of efficiency which points to better management of human resources but also that Governments may choose to disproportionately subsidize higher education or the resources are not reaching the poorest of the poor. In several countries, the per child spending also varies across regions. A good example of this is Indonesia where certain subnational governments are more affluent than others.[27]

Many LIC'S and lower middle-income countries (LMIC's) have not met international benchmarks for educational spending. In terms of both government expenditure and GDP proportions for education spending, 41 of 80 countries with available data met neither target.[28] It is estimated that LICs and LMICs would need to increase education spending by 4.2 to 6.5 percent of their GDP till 2030.[29] This is challenging due to the competing priorities that developing countries face from expanding healthcare access to increasing employment. It

becomes important for countries to mobilize more domestic resources to enhance government spending on education. According to IMF estimates, LICs and LMICs could boost their tax-to-GDP ratio by 9 percentage points by improving the design of their taxation system and stronger institutional frameworks.[30] This includes improving the design of core domestic taxes such as value added tax (VAT), corporate and personal income taxes. IMF also recommends broadening the tax base through neutral taxation of capital income and more efficient use of property taxes.[31] A case in point is China which since the early 1990s has modernized its tax administration. The system has 18 taxes with the revenue clearly divided between the central and local government, simplified and transparent revenue-share arrangements, and process automation. These changes reduced tax evasion and raised China's tax yield to nearly 20% of its GDP.[32]

Household expenditure towards education is a key component of the finance design in developing countries. Households contribute one-quarter of all global education expenditures, with the highest allocation being in LMIC's. In the poorest countries, households do not spend more than 1 percent of their budget on education.[33] In contrast, the priority assigned to education in LMIC's is higher, reaching a median of 2.9 percent.[34] This spending also differs across households depending on the size, income level, and the type of school children attend, with key differences between public and private education. Household spending in the past decade has increased in some countries such as India where it rose from 2.1 to 2.8 percent and decreased in several others including Montenegro, Namibia, and Colombia. This has been attributed in part to economic shocks of COVID-19.[35]

Alongside Government and household spending, international aid plays a critical role in financing public education in developing countries, where domestic resources are insufficient to meet learning needs. A common trend seen in education aid is its deprioritization on donors' list even though the volume of total education aid (ODA) has increased across some years; it was down to 7.6 percent in 2022.[36] Since 2018, the World Bank's disbursements of aid have increased, whereas aid from other larger donors, such as the United States and the United Kingdom have decreased. Across the largest donor countries, including Germany, France, the US, the UK, and Japan, education priorities also vary. For instance, 66% of the US's ODA was allocated to basic education, whereas Japan allocated 60% to post-secondary education.[37] LIC and LMIC's face an estimated US$97 billion annual financing gap to meet their Sustainable Development Goal 4 target of quality education by 2030. Cuts in funding, which are projected at $3.2 billion, would push the number of out-of-school children worldwide from 272 to 278 million.[38] These cuts would be felt deeply in contexts such as the Rohingya refugee population, where 350,000 children risk losing access to basic education.[39] It would also threaten other services such as meal programmes at schools and support for girls' education.

It is vital therefore to shift resources through multilateral channels, towards system strengthening, and towards lowering the cost of borrowing. What this does is increase the

flexibility of funding where funding can be pooled and allocated on actual system needs rather than narrow projects.[40] These recommendations also focus on improving governance, administrative and infrastructure within school districts, and effective teacher training. Finally, it reduces the burden of debt experienced by LIC and LMIC's through lowering costs and restructuring debt.[41]

COMPARITIVE CASE SELECTION: VIETNAM AND THE PHILIPPINES

To examine how education financing models influence equity in learning outcomes, this study compares Vietnam and the Philippines, two Southeast Asian countries that are relatively similar in population size and economic classification but have significant differences in educational outcomes. Both countries are classified as lower-middle-income countries according to the World Bank, share comparable regional contexts, and have undertaken several education reforms over the past two decades.[42]

Vietnam has a population of approximately 100 million, while the Philippines has around 115 million. The government expenditure on education as a % of total government expenditure for both countries is 15.4% and 16.5%, respectively.[43] Vietnam operates primarily through a centralized financing and governance model, where the central government maintains a strong control over budgeting, curriculum, and teacher deployment. Vietnam's strong results on international assessments of learning owe to the strong political commitment of its ruling party and high levels of societal engagement.[44] When PISA performances are compared between lower-income and higher-income OECD countries, Vietnam has been the only PISA outlier, with a performance on par with much wealthier countries. Although the country's performance in 2022 PISA assessment saw a decline, the research indicates that Vietnam's policies on investment in education and school infrastructure have enhanced access to pre-school for children.[45]

In the Philippines today, basic public education remains primarily the responsibility of the central government; however, within each province, city, and municipality, a local school board (LSB) is responsible for allocating and using government funds.[46] These exhibit elements of decentralized education financing that were intended to increase local responsiveness and efficiency. Despite this intention, schools have limited discretionary funding, and significant portions of the funds fail to reach schools, resulting in delays and worsening of school infrastructure.[47] Philippines ranked near bottom of the 81 countries participating in PISA, scoring significantly below the OECD average across all three subjects: math, reading, and science.[48] The learning poverty rate stands at 91% in Philippines, compared to 20% in Vietnam, presenting a stark difference in educational outcomes between the two countries.[49] By analyzing these two countries, this study seeks to answer how differences in education financing models explain divergent educational equity and learning outcomes.

VIETNAM: COUNTRY PROFILE

Vietnam's education system is managed by a strongly centralized governance structure under the administration of the Ministry of Education and Training (MOET).[50] Sub-national governments do play an important role in implementation. MOET sets guiding norms for local education policies and oversees an education system comprising more than 44,000 schools that serve 25 million students daily.[51] The Ministry of Finance and Ministry of Home affairs determine policies and the budgetary considerations relating to human resource. Although fiscal decentralization limits MOET's role, there are established "delegated policy objectives" regarding national assessment standards for schools, teachers, and students.[52] Compliance is ascertained through quarterly, bi-annual, annual, and five-year progress reports across criteria such as academic, organizational, and financial. School and teacher performance are rigorously evaluated through students' learning outcomes and performance on national assessments.[53]

At the local level, an established province's People's committee (PPC) determines education sector policy priorities and budgeting.[54] PPCs are expected to adhere to national guidelines and oversee policy implementation and education outcomes. The school standards focus on indicators such as infrastructure, equipment, organization, and staffing. Teachers' performance and learning outcomes are assessed through various performance categories, and both are actively monitored through the Bureau of Education and Training (BOET) and the Department of Education and Training (DOET).[55]

The education financing model uses a mix of general government budgets, private sources, and targeted funds for disadvantaged regions and communities. According to government policies, about 20 percent of annual budgetary expenses should be allocated for education, and a fiscal system has been established where tax revenues are redistributed to support poorer regions.[56] To further equity, the government ensures that teachers in disadvantaged regions are paid higher salaries than teachers in cities. In Vietnam, household and private spending are also important, and households contribute a larger share to education spending than the OECD average. This indicates strong cultural values towards education in Vietnamese families.[57]

In terms of allocation from the central government to provinces, it is based on transfer norms that are usually received by provinces as block grants over a three-year period.[58] If a province has a budget shortfall, this is accommodated by the Central government, although, this is only for the first year of the budget. For funds that are transferred, each school-age child receives different norms, based on the region they live. For instance, in the 2017-2020 period, the highest per-child fund was VND 5.05 million, which was two times higher than the lower per-child fund.[59] Finance departments at the province level prepare a province-wide proposed budget, which includes all decisions with respect to the allocation formulas for the schools' current expenditures. Although there is a set of national educational policies, provinces have a

significant degree of flexibility in their education finance. This results in significant variations from region to region in areas such as salaries and non-salary expenditures.

Beyond the financial architecture of Vietnam's education system, a series of policy reforms has played a critical role in shaping how resources are translated into learning outcomes. The political commitment and value attached to education are significant contributors to the government's concrete actions in the system. The Fundamental School Quality Level Program was established by MOET to develop minimum standards, review progress, and assess whether inputs lead to learning.[60] The central government has also established strong accountability mechanisms for school performance through a robust internal and external monitoring system. Over the past two decades, the government has placed an emphasis on preprimary education, as evidenced by mandating that every commune has one preschool, introducing bilingual education programs, and incentivizing the private sector to provide early childhood education (ECE). By 2017, the country provided preschool education to all five-year-old children.[61]

Additionally, the government has leveraged the Vietnamese people's deep respect for learning and teachers and has implemented policies to attract and support qualified teachers. Scholarships and tuition-free exemptions, rigorous evaluation and constructive feedback, professional development opportunities, and allowances are provided to help retain teachers.[62] According to the World Bank's analysis of PISA 2015, teacher quality measured by explanation and discussion of student questions is the most important factor contributing to student outcomes in Vietnam.[63]

From 1986 onwards, the Doi Moi reforms were initiated to transform Vietnam into a socialist, market-oriented economy. With respect to education, this included socializing education, meaning that it is everyone's responsibility to contribute to the education sector; access to quality education, including teacher investment and a focus on education for disadvantaged children; and strengthening vocational education.[64] Other key transformations included curriculum reform to make it more relevant to the requirements of a market economy and pedagogical improvements with a focus on an interdisciplinary perspective and a student-centered approach. The government also implemented the Education Management Information System (EMIS) for information storage, enabling them to draw conclusions about comparative outcomes across different regions and analyze demand for teacher-training.[65]

In conclusion, Vietnam presents a strong case, among other LMICs, for how a centralized governance with elements of a decentralized system, supports reasonably high outcomes with an equity focused orientation. It is important to note however, that despite its successes, the education system in Vietnam continues to suffer from challenges in improving secondary education enrollment, particularly for the socioeconomically disadvantaged. The country also faces regional disparities due to difficult transport conditions, internal migration, and high opportunity costs of schooling for young children willing to work.

PHILIPPINES: COUNTRY PROFILE

The governance structure of education systems in the Philippines is administered by four government agencies covering all levels of education: pre- and primary education, lower and upper secondary education, and alternative education programmes.[66] Local Government Units (LGUs) are responsible for the distribution of funds at the subnational level after receiving finance from the national government agencies. The General Appropriations Act (GAA) is an important legislation passed each year by the government to set annual expenditure limits. Under this legislation, education expenditures are divided into:

1. Salaries and other benefits
2. Maintenance and operating expenses
3. Capital outlays for construction projects.[67]

According to a new ruling in 2022, LGUs are allotted a higher share of national taxes (30-40%) each fiscal year, and the allocation is as follows: provinces (23%), cities (23%), municipalities (34%), and barangays, which are the smallest administrative division, receive 20%. The Ministry of Education provides funding for schools based on a funding formula that considers fixed expenses, per-student allocation, number of teachers, number of classrooms, and number of graduating students. However, this formula does not provide additional funding to schools in poorer regions or to marginalized groups, as in the Vietnamese case.[68] In terms of government expenditure, the education budget represented 16.4 of public expenditure in 2020. As per the Education 2030 Agenda, the minimum benchmark for public education finance was set to 4% of the GDP. The Philippines has never reached this benchmark, with the highest budget set in 2019 at 3.9 percent and the lowest expenditure in 2020 at 2.3 percent due to the impacts of the pandemic.[69] Within the education sector, primary education followed by lower secondary education receives the greatest share of investments. These investments align with the free K-12 program, where upper secondary education became a part of free and basic education.

Households in the Philippines bear high costs and have consistently spent more on education than the government, ranging from 52% in 2017 to 61.7% in 2020.[70] These expenditure items include allowances, transportation, and contributions to schools. The country also has a strong private education sector, which contributes significantly to high household expenses. Private schools account for about 40 per cent of all upper secondary schools nationwide.[71]

Over the past decade, the Philippines has introduced a series of reforms to improve basic education, access to primary education, and address inclusive education. These include the 2013 Basic Education Act, which increased the basic education cycle from 10 to 13 years. Significant improvements have been made to school infrastructure and teacher recruitment, including

reductions in student-teacher ratios. Through these reforms, poorer communities benefited the most, and the share of school-going children increased from 86 percent to 94 percent between 2002 and 2013.[72] The government also introduced the Indigenous Peoples Education Program which funds curriculum and resource development, and capacity building. The Madrasah Education program was implemented in schools with at least 15 Muslim students, and special education initiatives were introduced, including educational visits, camp activities, and sports activities.[73]

Despite these strides in education reform, there are significant accessibility issues in ECE across regions and income levels. Only 36% of barangays have at least one child development center (CDC). The ECE workforce is aging, which could lead to a shortage of experienced teachers in the future. Only 52% of educators have a college degree, and few have specialized training in ECE, in contrast to Vietnam's system, which emphasizes qualifications and rigorous evaluation.[74] In PISA 2022, 15-year-olds in the Philippines scored below the average in all areas: math, reading, and science. Only 16% of students attained Level 2 proficiency in mathematics, which is significantly lower than the OECD average (69%). Learning poverty in the Philippines is 56.4 percentage points higher than the average for the East Asia and Pacific region.[75] The combination of fiscal decentralization without strong equalization mechanisms and limited capacity at the LGU level continues to impede equitable education delivery. Regional funding disparities significantly affect student outcomes; regions with low per-student spending have poorer learning environments due to higher student-teacher and student-classroom ratios.

Research in the Philippines has shown that reducing the number of students per classroom is associated with better student outcomes. Schools in urban areas face congested learning environments, with 24 per cent of elementary and 30 per cent of high schools exceeding the Ministry of Education's student-classroom ratios.[76] Although spending on school infrastructure has improved, utilization rates for school infrastructure funds have been low over the past decade. Several infrastructure projects fail to be completed due to insufficient funds, delays in funding releases, and design issues.[77] Low levels of learning outcomes are also influenced by the quality of elementary and high-school teachers, which, according to the Primary Education Thinking Skills (PETS) assessments, is low in most subjects. This suggests that teachers face significant challenges in teaching the curriculum.

Data from the World Bank and the OECD reveal that the Philippines' education system suffers from significant inequities, with the poorest students, remote regions, and public schools consistently lagging in access, quality, and learning achievement. Lower levels of funding are a key reason schools serving poorer communities tend to have lower ratings for the implementation of school-based management – an approach to education governance in which decision-making authority rests with individual schools.While allocation systems have managed to target funding to certain schools, there are many schools that miss out on additional supports and well-off schools that continue to receive support.[78] Additionally, there are delays in identifying school needs, which disrupts schools' operations and learning for students.

LESSONS AND CROSS-COUNTRY INSIGHTS

The comparison between Vietnam and the Philippines demonstrates that the structure and governance of education financing shape equity and learning outcomes. Although both countries are lower-middle-income economies with comparable population sizes, their education outcomes diverge due to differences in policy design, mechanisms of allocating resources across regions and students, and institutional capacity.

1) Centralized Redistribution Can Strengthen Equity When Effectively Managed

   Vietnam's centralized system, administered by MOET and supported by strong accountability mechanisms, has enabled a more coherent approach to redistributing funds in disadvantaged regions. By contrast, the Philippines' somewhat decentralized model does not allow for strong equalization mechanisms to compensate for disparities in local fiscal capacity. In recent years, local government funding has been declining, and funding from parent-teacher associations has also been negligible. Although allocation systems in the Philippines can target funds, several needy schools are missing out on additional support. Furthermore, there are significant disparities across regions, as wealthier LGUs can invest more in school infrastructure and teacher deployment. Improvements in allocation mechanisms can be achieved through better planning. For instance, school infrastructure improvements can be developed over a two- or three-year plan, enabling the Philippines' Department of Education to monitor projects effectively and improve forecasting of school improvement projects.[79]

   School allocation decisions should also see better input from division offices and schools, as funding decisions taken at the central level often do not account for local school conditions. These insights suggest that effective vertical equity requires autonomy at the subnational level, along with strong central oversight and redistributive formulas that account for diverse student needs.

2) Teacher Quality and Deployment Policies Matter as Much as Financial Inputs

   Vietnam's emphasis on teacher professionalism, funding for continuous professional development, and incentives for teachers in disadvantaged areas has contributed directly to stronger student outcomes. Teachers are evaluated rigorously and supported through allowances and strong feedback systems. The Philippines, however, faces teacher shortages across several regions, uneven distribution of teachers, and limited opportunities for professional growth and specialized training. Research indicates that vulnerable groups are more likely to complete schooling when teachers have higher levels of education.[80] One of the most important variables related to teaching is content knowledge. Teacher content knowledge is one of the largest predictors of student test scores. Related to this, the quality of teacher certifications for identifying "excellent

teachers" has also been correlated with higher learning outcomes.[81] Vietnam has been active in supporting teacher policies, particularly in promoting higher qualifications and teacher training. Teachers in the Philippines face a shortage of master teacher positions, and only a limited percentage are promoted to them, limiting promotion opportunities. These learnings tell us that strengthening teacher quality through recruitment, training, rigorous monitoring, and incentives is crucial to translating financial resources into improved student learning.

3) Address Funding and Quality Inequalities

Both countries allocate a similar share of government expenditures to education, yet Vietnam achieves far better learning outcomes. This difference reflects the efficiency and targeting of funds rather than their absolute size. Vietnam's block-grant transfer norms, weighted per-child allocations, and targeted funds for ethnic minorities create a financial framework that directs resources towards minority groups. Although Vietnam continues to display large gaps in scores between urban and rural/remote areas, and between the poor and wealthy, these gaps have been reduced, with scores increasing across all three groups largely due to these funding mechanisms. In contrast, the Philippines' education system has a funding formula that lacks the equity component. This formula does not account for different levels of socioeconomic disadvantage, which leads households to bear a significant proportion of total education costs. Poorer students not only attend high schools with teachers with limited content knowledge but also attend schools with lower funding and resources. By introducing an equity component into their funding formula, the Philippines can enable more equalized school funding, which would provide schools in poorer areas with additional resources to support the learning needs of minority groups.[82] Addressing funding through this approach depicts that equity gains depend not only on spending more, but on spending strategically through weighted student formulas and targeted interventions.

4) Policy Coherence is a Critical Differentiator

Vietnam's policy reforms, including the Fundamental School Quality Level Program, early childhood education, school management, and curriculum modernization, are aligned with the country's long-term national priorities. Although the Philippines' reforms, including K-12 expansion and reforms for the indigenous people's education program, often see uneven implementation across regions and are constrained by local capacity and funding. Sustained improvements require coherent reforms rather than fragmented initiatives.

This cross-country comparison shows that countries with limited resources can achieve strong educational outcomes if they prioritize equity in funding formulas and allocations, teacher quality, and overall policy coherence.

POLICY RECOMMENDATIONS

The comparative analysis of Vietnam and the Philippines underscores that improving educational equity in low and middle-income countries requires policy reforms that address the volume of financing as well as the governance structures, allocation methods, and institutional capacities that determine how effectively financing is translated into learning outcomes. Based on the evidence reviewed, the following policy recommendations aim to strengthen vertical equity and enhance school and system efficiency.

Recommendation	Key Actions
1. Strengthen allocation mechanisms and formula-based funding models	• Adopt or expand weighted student funding formulas that allocate more resources to schools in disadvantaged and remote regions. • Create equity weights for poverty, geographical location, and minority status.[83] • Ensuring that budget preparation at the school and national level requires the establishment of well-defined goals and outcomes, as well as determining priorities and allocating funding to high-priority items. • Establishing transparent monitoring mechanisms through an Education Management Information System (EMIS) reporting system.[84]
2. Improving teacher recruitment, deployment, and teacher support system	• Developing guidelines/norms for school-level teaching needs and for subject-specific teachers in high schools • Implement structured, mandatory professional development programs that are aligned to student's learning outcomes. • Increasing authority of schools and school division offices to deal with issues of absenteeism among teachers and link teachers' attendance to their career development and performance bonuses.[85] • Strengthen teacher assignment algorithms and revise teacher transfer policies to improve teacher distribution and reduce issues related to school needs and teacher supply.[86]
3. Enhance Accountability and Monitoring Systems	• Expand and integrate EMIS to enable real-time monitoring of school resources and fund utilization. • For infrastructure related projects, project selection criteria and feedback/grievance

	mechanisms can be established to increase transparency. • Enable schools to monitor projects more closely and provide feedback for all stakeholders to use. • Establishing school resource inventory modules that monitor availability of learning materials and infrastructure such as classrooms. This includes mapping school resources and tagging them by age and condition. This will allow for better prioritization of investment for under-resourced schools.[87]
4. Introducing conditional cash transfers for vulnerable students	• High household expenditure, increased tuition fees at high schools, and opportunity costs of education in lower-income countries tells us of the scope in using CCT's in alleviating this problem. • This includes targeting beneficiaries using poverty indicators, school context, geographical location, among others.[88] • Minimum attendance and progression benchmarks can be established to set clear education linked conditions. • Strengthening coordination with schools and local division offices to identify at-risk populations and provide support.
Strengthening School-Based Management	• Providing enhanced training to school principals and parent-teacher associations on school-based management that enable them to work together and lead the preparation of school development plans. • Provide additional support and expertise on school-based management to schools who serve disadvantaged students. • Enforcing regulations to make information publicly available on school performance and use of funds.[89]

5. Expanding Early Childhood and Complementary school Services	• Increasing public investments to build and staff ECE centers, especially in rural and poor regions. • Implementing standardized ECE teacher qualification pathways. • Establishing rigorous health checks and nutrition interventions that have the potential to improve education outcomes when started in early childhood. • Improving coordination among agencies such as DepEd, National Nutrition Council, ECCD council, and LGU's to enhance access and effectiveness of complementary interventions.[90]
6. Leverage partnerships to create strengthened learning-oriented systems. According to the Independent Evaluation Group of the World Bank, partnerships with international organizations such as the World Bank have been key to delivering strategic improvements in systems and outcomes across countries like Vietnam and Brazil.[91]	• Establishing education projects and Country Partnership Frameworks (CPF's) that include indicators for learning improvements. • Partnering with Department of Education's in countries to improve national education assessment capabilities and systems for better data decision-making.

REFERENCES

[1] "What You Need to Know about Education Financing." UNESCO, July 28, 2025. https://www.unesco.org/en/education-financing/need-know.
[2] Berne, Robert, and Leanna Stiefel. "Measuring Equity at the School Level: The Finance Perspective." *Educational Evaluation and Policy Analysis* 16, no. 4 (December 1994): 405–21. https://doi.org/10.3102/01623737016004405.
[3] What You Need to Know about Education Financing." UNESCO, July 28, 2025. https://www.unesco.org/en/education-financing/need-know.
[4] Ibid
[5] ibid
[6] Ibid
[7] Ibid
[8] ibid
[9] Berne, Robert, and Leanna Stiefel. "Measuring Equity at the School Level: The Finance Perspective.
[10] Ibid, 406
[11] Ibid, 407
[12] Ibid, 407
[13] Ibid, 407
[14] "How Money Matters: Education Funding and Student Outcomes[Fact Sheet]." Learning Policy Institute, April 8, 2025. https://learningpolicyinstitute.org/product/how-money-matters-factsheet.
[15] "Education Finance." World Bank, April 14, 2025.
[16] Patrinos, Harry, and Nobuyaki Tanaka. "Education: Innovative Financing in Developing Countries." World Bank, April 2024. https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099042624181519346/p5009781490d0b070195ed1cd5436f2922c.
[17] Patrinos, Harry, and Nobuyaki Tanaka. "Education: Innovative Financing in Developing Countries
[18] ECD (2017), The Funding of School Education: Connecting Resources and Learning, OECD Publishing, Paris. http://dx.doi.org/10.1787/9789264276147-en
[19] ECD (2017), The Funding of School Education: Connecting Resources and Learning,
[20] Ibid, 21
[21] Ibid, 21
[22] Ibid, 22
[23] Ibid, 22
[24] Tanaka, Nobuyuki; Angel-Urdinola, Diego; Bend, May; Hu, Yitong; Rivera-Olvera, Angelica; Antoninis, Manos; Murakami, Yuki; Montoya, Silvia. *Education Finance Watch 2024 (English).* Washington, D.C. : World Bank Group. http://documents.worldbank.org/curated/en/099102824144527868
[25] Murti, Mamta. "Quality Education Requires Greater and More Efficient Financing." World Bank Blogs, June 16, 2023. https://blogs.worldbank.org/en/education/quality-education-requires-greater-and-more-efficient-financing.
[26] Murti, Mamta. "Quality Education Requires Greater and More Efficient Financing.
[27] ibid
[28] Tanaka, Nobuyuki, et al, *Education Finance Watch 2024, 8*
[29] Ibid,9
[30] Gaspar, Vitor, Mario Mansour, and Charles Vellutini. "Countries Can Tap Tax Potential to Finance Development Goals." IMF, September 19, 2023. https://www.imf.org/en/Blogs/Articles/2023/09/19/countries-can-tap-tax-potential-to-finance-development-goals.

[31] Gaspar, et al. Countries Can Tap Tax Potential to Finance Development Goals.
[32] Brondolo, John David, and Zhiyong Zhang. Tax administration reform in China: Achievements, challenges, and reform priorities, March 17, 2016. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2844125.
[33] Tanaka, Nobuyuki, et al, *Education Finance Watch 2024, 12*
[34] Ibid, 12
[35] Ibid, 13
[36] Ibid, 10
[37] Ibid, 11

[38] "Funding Cuts Could Push 6 million More Children out of School, Warns UNICEF | UN News." United Nations News, September 2, 2025. https://news.un.org/en/story/2025/09/1165764.

[39] Funding Cuts Could Push 6 million More Children out of School, Warns UNICEF. UN News.
[40] "UNESCO Global Education Monitoring Report: Leadership in Education, UNESCO, 2025. https://www.european-agency.org/news/2024-gem-report.

[41] UNESCO Global Education Monitoring Report: Leadership in Education.
[42] "Data for Philippines, Lower Middle Income, Viet Nam." World Bank Open Data. Accessed October 23, 2025. https://data.worldbank.org/?locations=PH-XN-VN.
[43] "Data for Philippines, Lower Middle Income, Viet Nam." World Bank Open Data.

[44] London, Jonathan. "Vietnam's Education System: How Coherent Is It for Learning?" Rise Programme, February 2023. https://riseprogramme.org/sites/default/files/2023-03/Vietnam's Education System - How Coherent is it for Learning_0.pdf.

[45] Kataoka, Sachiko, Le An Vinh, Sandhya Kitchlu, and Keiko Inoue. "Vietnam's Human Capital: Education Success & Future Challenges." Human Capital Data Portal, August 2020. https://documents1.worldbank.org/curated/en/261401596433155112/pdf/Vietnam-s-Human-Capital-Education-Success-and-Future-Challenges.pdf.

[46] Llego, Mark Anthony. "Analyzing the Philippine Education System: Challenges, Reforms, and Future Directions - Teacherph." TEACHERPH, June 2024. https://www.teacherph.com/philippine-education-system-challenges-reforms-analysis/.

[47] Llego, Mark Anthony. "Analyzing the Philippine Education System
[48]Samarrai, Samer Al, Vandana Rao, Jason Alinsunurin, and Himdat Bayusuf. "Assessing Basic Education Service Delivery in the Philippines: Public Education Expenditure Tracking and Quantitative Service Delivery Study." World Bank, June 17, 2016. https://www.worldbank.org/en/country/philippines/publication/assessing-basic-education-service-delivery-in-the-philippines-public-education-expenditure-tracking-and-quantitative-service-delivery-study.
[49] "Philippines Learning Poverty Brief June 2022." Edu Analytics, 2022. https://documents1.worldbank.org/curated/en/099000207152223103/pdf/IDU002b5536c0db4104ec3087d809906ec2eae56.pdf.
[50] London, Jonathan. "Vietnam's Education System: How Coherent Is It for Learning, 16
[51] Ibid, 17
[52] Ibid, 18
[53] Kataoka, Sachiko, et al. "Vietnam's Human Capital: Education Success & Future Challenges, 17
[54] [54] London, Jonathan. "Vietnam's Education System: How Coherent Is It for Learning, 17
[55] Ibid, 18
[56] Kataoka, Sachiko, et al. "Vietnam's Human Capital: Education Success & Future Challenges, 14
[57] Ibid, 15
[58] London, Jonathan. "Vietnam's Education System: How Coherent Is It for Learning, 20

[59] Ibid, 20
[60] Ibid, 22
[61] Kataoka, Sachiko, et al. "Vietnam's Human Capital: Education Success & Future Challenges, 18
[62] Ibid, 16
[63] Parandekar, Suhas D., and Elizabeth K. Sedmik. "Unraveling a Secret: Vietnam's Outstanding Performance on the Pisa Test." World Bank, April 2016. https://documents.worldbank.org/en/publication/documents-reports/documentdetail/258431468196137980/unraveling-a-secret-vietnams-outstanding-performance-on-the-pisa-test.
[64] Kataoka, Sachiko, et al. "Vietnam's Human Capital: Education Success & Future Challenges, 23
[65] Ibid, 33

[66]. Tenazas, Nicholas Mendoza. "The Financing of Education in Philippines." UNESCO Digital Library, 2022. https://unesdoc.unesco.org/ark:/48223/pf0000384382?posInSet=150&queryId=5ec34064-c34a-4442-8547-6b9b0da251df.
[67] Tenazas, Nicholas Mendoza. "The Financing of Education in Philippines, 3
[68] "Financing for Equity." Philippines | FINANCING FOR EQUITY | Education Profiles, 2024. https://education-profiles.org/eastern-and-south-eastern-asia/philippines/~financing-for-equity.
[69] Financing for Equity." Philippines
[70] Tenazas, Nicholas Mendoza. "The Financing of Education in Philippines, 7
[71] Ibid, 8
[72] Samarrai, Samer Al, et al. "Assessing Basic Education Service Delivery in the Philippines", 14
[73] Financing for Equity." Philippines | FINANCING FOR EQUITY | Education Profiles, 2024
[74] Llego, Mark Anthony. "Analyzing the Philippine Education System"
[75] Philippines Learning Poverty Brief June 2022." Edu Analytics, 2022
[76] Samarrai, Samer Al, et al. "Assessing Basic Education Service Delivery in the Philippines", 29
[77] Ibid, 21
[78] Ibid, 31

[79] World Bank. "Philippines - Infrastructure for Safer and Resilient Schools Project." World Bank, October 6, 2024. https://documents.worldbank.org/pt/publication/documents-reports/documentdetail/099061024172038738/099061024172038738.

[80] Kataoka, Sachiko, et al. "Vietnam's Human Capital: Education Success & Future Challenges, 16
[81] Ibid, 33

[82] "Philippines Basic Education Public Expenditure Review." World Bank, October 1, 2020. https://openknowledge.worldbank.org/server/api/core/bitstreams/3d726a60-03c8-553f-8de8-214e29234b45/content.

[83] Financing for Equity." Philippines | FINANCING FOR EQUITY | Education Profiles, 2024. https://education-profiles.org/eastern-and-south-eastern-asia/philippines/~financing-for-equity.

[84] Kataoka, Sachiko, et al. "Vietnam's Human Capital: Education Success & Future Challenges, 33

[85] London, Jonathan. "Vietnam's Education System: How Coherent Is It for Learning,

[86] Arias, Omar, Veronica Diaz, and Marcela Bernal. "Scaling and Sustaining Teacher Policies for Educational Transformation." World Bank Blogs, March 27, 2024. https://blogs.worldbank.org/en/education/Scaling-and-sustaining-teacher-policies-for-educational-transformation.

[87] Beemer, Sandra F., Binh Thanh Vu, Sally Brinkman, Wendy Jarvie, and Kevin Macdonald. "Implementing and Evaluating Interventions to Improve School Readiness and Early Literacy: Experience from the Pacific Early Age Readiness and Learning Program (Vol. 1 of 2) : Operational Guide." World Bank, June 2019. https://documents.worldbank.org/en/publication/documents-reports/documentdetail/981801563250242375/operational-guide.

[88] Samarrai, Samer Al, et al. "Assessing Basic Education Service Delivery in the Philippines
[89] Ibid, 37
[90] Ibid, 35

[91] "Learning in World Bank Lending - an Independent Evaluation." Independent Evaluation Group, March 24, 2025. https://ieg.worldbankgroup.org/evaluations/learning-world-bank-lending.